25901821 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

SECOND SEMESTER

Skill Development Components (SDC) - MG2SDCMFT100 - HUMAN BODY: STRUCTURE, FUNCTION & ENERGY METABOLISM

(2025 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer any **10** questions

Each question carries **1** marks

1. Which of the following is a flat bone? [K] / [CO1]

   a). Femur b). Humerus c). Sternum d). Tibia

2. Dehydration can negatively affect exercise performance by: [U] / [CO2, CO5]

   a). Increasing joint flexibility b). Reducing blood volume and thermoregulation c). Enhancing VO_2 Max d). Increasing muscle glycogen

3. Understanding joint kinetics helps coaches and athletes to: [A] / [CO3, CO5]

   a). Reduce injury risk and improve performance b). Increase fat accumulation exclusively c). Slow neural response d). Replace hydration needs

4. Understanding kinematic chains helps athletes and coaches to: [A] / [CO3, CO5]

   a). Design exercises that enhance b). Reduce energy system activation c). Slow heart rate permanently d). Store fat exclusively

performance, efficiency, and injury prevention

5. Understanding anaerobic adaptations helps athletes to: [A] / [CO2, CO4, CO5]

a). Improve explosive power, strength, and performance while reducing injury risk | b). Reduce oxygen delivery permanently | c). Increase fat storage exclusively | d). Slow cardiovascular adaptation

6. Proper warm-up before sports improves joint function by: [U] / [CO2, CO5]

a). Reducing synovial fluid | b). Increasing joint lubrication and flexibility | c). Decreasing blood flow | d). Strengthening bones immediately

7. Which hormone is primarily responsible for increasing blood glucose levels during exercise? [K] / [CO1, CO2, CO4]

a). Insulin | b). Glucagon | c). Cortisol | d). Thyroxine

8. Which of the following is a key application of biomechanics in sports? [U] / [CO3, CO5]

a). Designing effective training techniques | b). Digesting food faster | c). Slowing heart rate | d). Increasing fat storage

9. Kinesiology is primarily the study of: [K] / [CO3]

a). Hormonal responses to exercise | b). Human movement, including muscles, joints, and biomechanics | c). Lung volumes and respiration | d). Blood circulation

10. Which cardiovascular adaptation is commonly seen after prolonged aerobic training? [U] / [CO2, CO4]

a). Lower resting heart rate and increased cardiac output | b). Increased resting heart rate | c). Decreased stroke volume | d). Reduced blood volume

11. The knee-jerk reflex is an example of a: [K] / [CO1, CO2]

a). Polysynaptic reflex | b). Monosynaptic reflex | c). Conditioned reflex | d). Voluntary action

12. The basic structural and functional unit of the human body is the: [K] / [CO1]

a). Atom b). Cell c). Tissue d). Organ

[1x10 = 10]

Part B

short answer

Answer any **5** questions

Each question carries **2** marks

13. Why is knowledge of center of gravity, balance, and stability important for injury prevention in athletes? [U] / [CO3, CO5]

14. How does understanding kinesiology improve performance in lifting, running, and throwing? [U] / [CO3, CO5]

15. How can understanding fatigue mechanisms help athletes prevent injury during high-intensity training? [A] / [CO2, CO3, CO4, CO5]

16. Why is the nervous system important for coordination and performance in sports? [U] / [CO1, CO2, CO3, CO5]

17. Name any two lung volumes involved in breathing. [K] / [CO1, CO2]

18. How does biomechanical analysis improve performance in athletes? [U] / [CO3, CO5]

19. What is an antagonist muscle and what role does it play in movement? [U] / [CO3, CO5]

[2x5 = 10]

Part C

short Essay

Answer any **4** questions

Each question carries **5** marks

20. Explain the role of macronutrients (carbohydrates, proteins, fats) in supporting energy production during exercise [U] / [CO2, CO4]

21. How can understanding the major muscle groups and their functions help in applying correct biomechanics to prevent injuries during exercise [A] / [CO1, CO2, CO3, CO5]

22. Explain how blood circulation changes during exercise to meet the increased oxygen and nutrient demands of muscles [U] / [CO1, CO2, CO4]

23. How can understanding force, motion, and torque help athletes improve efficiency and prevent injuries during training or competition [A] / [CO3, CO5]

24. How can understanding human motion help in enhancing sports performance and preventing injuries [A] / [CO1, CO2, CO3, CO5]

25. How can understanding fuel utilization at different exercise intensities help athletes improve performance and recovery? [A] / [CO2, CO4, CO5]

[5x4 = 20]

Part D

Essay

Answer any **2** questions

Each question carries **15** marks

26. Explain how posture and gait affect movement efficiency and the risk of musculoskeletal injuries. Provide examples from sports or daily activities [A] / [CO1, CO3, CO5]

27. Describe how knowledge of energy systems can help athletes plan training, optimize performance, and delay fatigue during exercise [A] / [CO2, CO4, CO5]

28. Explain the three classes of levers in the human body with examples for each. Discuss how understanding lever mechanics helps in analyzing movement efficiency [U] / [CO1, CO3, CO5]

[15x2 = 30]